Stretch-and-Squeeze Physics:

A Methodological Discipline for Revealing Hidden Structure Through Representational Choice

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Abstract

Einstein's most profound contributions were not new entities but new representations: special relativity dissolved absolute rest by squeezing physics to observer-coupled measurements; general relativity dissolved the gravity-inertia distinction by recognizing both as artifacts of coordinate choice. This paper extracts the underlying method—*stretch-and-squeeze transformations*—and proposes it as an explicit discipline for contemporary foundational physics. The method proceeds by first exposing the full gauge freedom of a formulation (stretch), then compressing to a new slice aligned with operational observables (squeeze). We demonstrate the technique through two worked examples in gravitational physics: a tidal-field-first formulation that foregrounds what detectors like LIGO actually measure, and a boundary-first formulation that explains why asymptotic structure (BMS symmetries, soft theorems) was discovered at null infinity rather than in the bulk. We conclude with a speculative application: the dark matter problem may be partly representational, with galactic dynamics described in a frame that makes certain regularities appear anomalous when they might be natural in a curvature-primitive or acceleration-first description. The core claim is methodological: when phenomena appear elusive, the first intervention should be representational, not ontological.

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1 Introduction: Einstein's Method, Extracted

1.1 Two Dissolutions

In 1905, physicists were searching for the ether—the medium through which light supposedly propagated. The Michelson-Morley experiment had failed to detect Earth's motion through this medium, generating a puzzle: why couldn't we measure absolute velocity? The instinct was to add structure: Lorentz proposed length contraction as a dynamical effect, FitzGerald proposed similar mechanisms. The ether was there; we just couldn't see it properly.

Einstein's move was different. He noticed that "velocity relative to the ether" was not a quantity that appeared in any actual measurement. What appeared in measurements were clock readings and rod lengths—observer-coupled quantities. By promoting these operational observables to primitive status and demoting absolute rest to a derived (and, it turned out, unnecessary) concept, the puzzle dissolved. There was no ether to detect because "absolute rest" was a representational artifact, not a physical fact.

A decade later, Einstein performed the same maneuver on gravity. The equivalence principle—the observation that free-fall is locally indistinguishable from the absence of gravity—suggested that gravitational "force" might be another representational artifact. An observer in a falling elevator and an observer in deep space have the same local physics. The difference between "gravity present" and "gravity absent" was a coordinate choice, not a physical distinction. Gravity was not a force to be explained but a feature of how we parameterize spacetime.

Both breakthroughs share a structure:

1. A phenomenon appeared puzzling or elusive in the standard representation.
2. The standard representation contained implicit commitments (absolute rest, gravitational force) that were not operationally grounded.
3. Promoting operationally-coupled variables to primitive status dissolved the puzzle.
4. No new entities were added; structure was *subtracted* by recognizing representational artifacts.

This paper argues that this pattern—which we call *stretch-and-squeeze*—deserves explicit methodological status. When a phenomenon resists explanation, the default scientific instinct is to add structure: new particles, new fields, new symmetries, new dimensions. This instinct is often correct. But there is a logically prior question that is frequently skipped: *Is the elusiveness itself an artifact of representational choice?*

1.2 The Cognitive Bias We Are Correcting

There is an asymmetry in how physicists respond to anomalies. Adding hypotheses feels like progress—you have proposed something, you can test it, you can be wrong in a productive way. Questioning the coordinate system feels like retreat—surely we chose standard variables for good reasons?

But representational choices accumulate. A field inherits a canonical formulation from its founders; students learn that formulation as "the physics"; alternatives are relegated to footnotes about gauge freedom that everyone acknowledges and no one systematically exploits. Over time, the representation becomes invisible infrastructure, and its implicit commitments become available for mistaking as physical facts.

The discipline we propose is simple: *before adding ontology, audit your representation.* Check whether your primitive variables couple to physical interfaces. Check whether your "elusive" degrees of freedom might be gauge artifacts. Check whether a different slice through the same physics would make manifest what currently appears hidden.

1.3 Scope and Stakes

This paper does not propose new physics. It proposes a methodological habit that is undersupplied in foundational research. We illustrate the method with two worked examples from gravitational physics where the technique has already paid off (even when not explicitly named as such), and we conclude with a speculative application to the dark matter problem—not to claim a solution, but to demonstrate that the representational question has been underexplored.

The stakes are straightforward: if even one major outstanding puzzle is partly representational, the return on this methodological investment is enormous. And the history of physics suggests this is not unlikely.

2 The Method: Stretch-and-Squeeze Transformations

2.1 Overview

A *stretch-and-squeeze transformation* is a structured change of representation designed to expose operationally-relevant degrees of freedom that may be obscured in a canonical formulation. The procedure has two phases:

1. **Stretch:** Expose the full gauge orbit or mathematical redundancy in the current formulation. This means explicitly identifying which features of the description are physical and which are artifacts of representational choice. The stretch phase prevents premature commitment to a particular gauge or coordinate system.

2. **Squeeze:** Select a new slice through the gauge orbit that aligns with operational observables—quantities that directly couple to measurement interfaces. This compresses the description around degrees of freedom that are both physical and accessible.

The key insight is that "operational observable" is not a fixed category. Different experimental arrangements couple to different aspects of the physics. A representation optimized for one interface may obscure structure that is manifest from another.

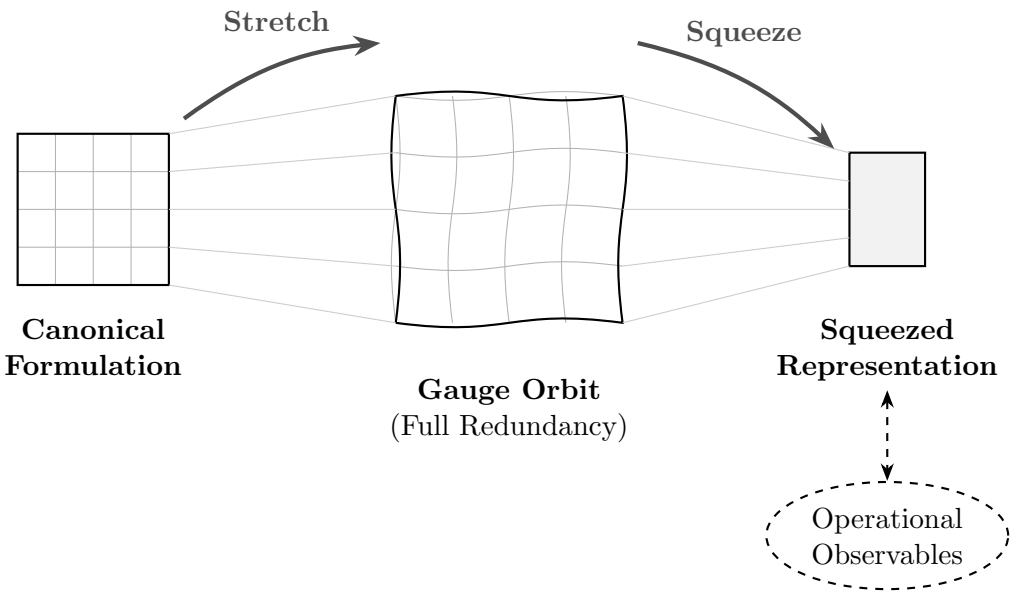

Figure 1: **The Stretch-and-Squeeze Method.** The process involves first exposing the full mathematical redundancy of a formulation (stretching), then projecting down to a slice that aligns with physical measurement interfaces (squeezing).

2.2 Diagnostic Checklist

When confronting an elusive phenomenon, run the following audit:

Representational Audit Protocol

1. **Identify your primitives.** What variables does your formulation treat as fundamental? (Fields, metrics, potentials, particle positions?)
2. **Check interface coupling.** Do your primitive variables couple directly to a physical measurement interface? Or do observables require integrating over auxiliary structures that cancel out?
3. **Expose your gauge.** What gauge or coordinate choice have you implicitly fixed? What degrees of freedom did that fixing hide?
4. **Invert the hierarchy.** If you promoted a currently-derived quantity to primitive status, what would become manifest? What would become obscured?
5. **Relocate the observer.** Are your "elusive" degrees of freedom artifacts of a bulk description that would be natural at a boundary (spatial infinity, null infinity, a horizon)?
6. **Test for frame dependence.** Is the quantity you cannot find (a particle, a field value, a force) frame-invariant? Or does it exist in some frames and not others?

2.3 Failure Modes and Antipatterns

You may be in a suboptimal representation if:

- Your observables require reconstructing physical quantities through a detour (e.g., computing a potential and then differentiating twice to get a tidal field that you could have measured directly).
- Your "fundamental" variables transform nontrivially under symmetries that leave all measurements invariant.
- Particle number is your primitive, but your detector doesn't count particles—it measures field correlators or response functions.
- You are searching for an entity (the ether, a local graviton, absolute simultaneity) that no operational procedure could detect.
- A quantity appears fine-tuned or anomalous in your coordinates but is natural (order-unity, symmetry-dictated) in another parameterization.

2.4 What Survives the Transformation

A stretch-and-squeeze transformation preserves physical content while reallocating conceptual structure. But what exactly is "physical content"?

The invariants that survive representational change form a hierarchy:

1. **Asymptotic data:** S-matrix elements, scattering amplitudes, memory effects—quantities defined at boundaries where gauge freedom is exhausted.
2. **Curvature invariants:** Scalar contractions of the Riemann tensor, proper times along worldlines, geodesic deviation—quantities that do not depend on coordinate choice.
3. **Detector response functions:** What a specified apparatus would actually register—correlators evaluated at the apparatus location.
4. **Relational observables:** Coincidences, correlations, ratios—quantities defined by relations between physical systems rather than relations to a coordinate grid.

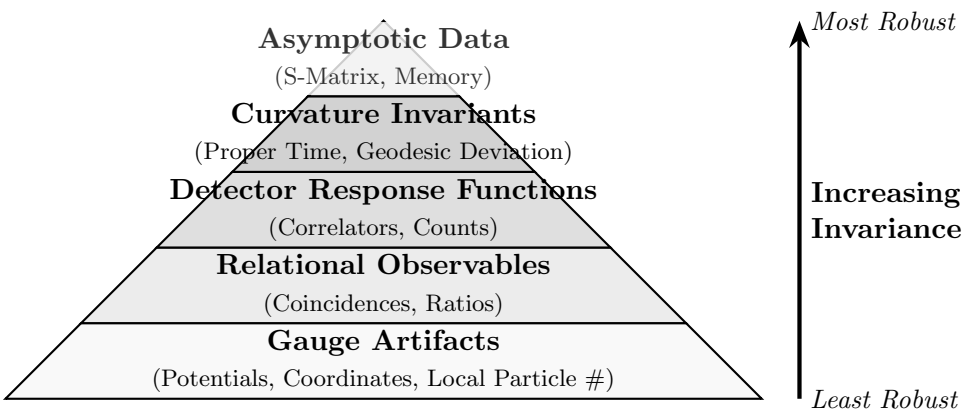

Figure 2: **Hierarchy of Invariants.** The method prioritizes quantities near the top of the pyramid, which are invariant across representational choices, over those at the bottom, which are often artifacts of specific frames or gauges.

Quantities that are *not* invariant include: local particle number (frame-dependent, as Unruh showed), field values at a coordinate point (gauge-dependent), gravitational "force" at a location (removable by free fall), and potential values (only differences are physical).

The methodological upshot: anchor your analysis on invariants from this hierarchy, and treat other quantities as representational conveniences that may or may not be well-suited to a given problem.

2.5 Comparison of Representations

Table 1 illustrates how the same gravitational physics maps across canonical and squeezed representations.

Table 1: Canonical vs. Observer-Coupled Representations of Gravity

Component	**Canonical (Metric-First)**	**Squeezed (Curvature-First)**
Primitive Variable	Metric $g_{\mu\nu}$	Tidal tensor E_{ij} or Weyl scalar Ψ_4
Derived Quantity	Curvature (two derivatives of g)	Metric (double integration)
Gauge Content	Diffeomorphism freedom in g	Eliminated at observer worldline
Detector Coupling	Indirect (via geodesic equation)	Direct (geodesic deviation)
Particle Notion	Graviton $h_{\mu\nu}$ in perturbation theory	Curvature correlator $\langle EE \rangle$
Natural Domain	Bulk spacetime	Observer frame or $\mathcal{I}^+$

3 Example I: Tidal-Field-First Gravity

Stakes: *This is how LIGO actually works—and why "gravitational wave detection" is already an implicit squeeze to observer-coupled variables, even if not always framed that way.*

3.1 The Standard Story and Its Detour

In the canonical formulation, gravitational waves are perturbations $h_{\mu\nu}$ of the spacetime metric. One linearizes Einstein's equations, fixes a gauge (typically transverse-traceless), and interprets the resulting wave equation as describing propagating spin-2 degrees of freedom—gravitons, in the quantum version.

This is mathematically correct. But consider what LIGO actually measures: the differential change in length between its two arms. This is not directly $h_{\mu\nu}$; it is the relative acceleration of test masses, which is governed by the *tidal field*—the Riemann curvature component R_{0i0j} in the detector's frame.

The standard analysis gets to the right answer, but through a detour: start with the metric perturbation, differentiate twice, extract the tidal effect, predict the arm-length change. The tidal-field-first formulation inverts this: start with the operational observable (relative acceleration), treat it as primitive, and regard the metric as a derived potential.

3.2 Geodesic Deviation as the Primitive Equation

Consider a family of nearby free-falling worldlines with separation vector ξ^μ. The geodesic deviation equation states:

$$\frac{D^2\xi^\mu}{D\tau^2} = -\, R^\mu{}_{\nu\rho\sigma}\, u^\nu \xi^\rho u^\sigma \tag{1}$$

where u^μ is the four-velocity. In the observer's local orthonormal frame, define the *electric* part of the Weyl tensor (which equals the Riemann tensor in vacuum):

$$E_{ij} \equiv R_{0i0j}. \tag{2}$$

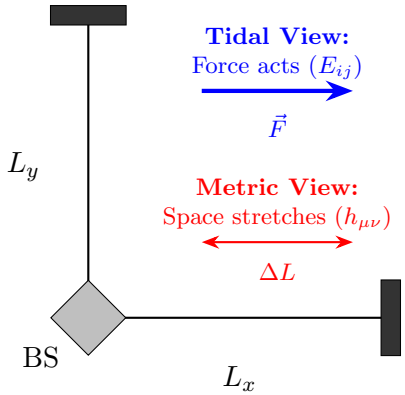

Figure 3: **LIGO: Two Perspectives.** In the canonical view (red), the gravitational wave is a metric perturbation that stretches the coordinate grid. In the squeezed view (blue), the wave is a tidal force field that accelerates the test masses relative to the central beamsplitter.

This tidal tensor directly encodes the relative accelerations among test masses—exactly what a gravitational wave detector measures.

3.3 The Metric as Derived Potential

In the weak-field, slow-motion limit, the tidal tensor is related to the transverse-traceless metric perturbation by:

$$E_{ij}(\tau) \approx -\frac{1}{2}\frac{d^2}{d\tau^2}h_{ij}^{TT}. \tag{3}$$

From the squeezed perspective, this equation is read right-to-left: the metric perturbation h_{ij}^{TT} is a double integral of the physically-primitive tidal field. The metric is a *potential* for the curvature, just as the vector potential A_μ is a potential for the electromagnetic field strength $F_{\mu\nu}$.

This is not merely a notational preference. The gauge freedom in $h_{\mu\nu}$ (diffeomorphisms, choice of TT frame) is entirely absorbed when we work with E_{ij} evaluated on a specific observer's worldline. The squeezed representation has fewer unphysical degrees of freedom precisely because it is anchored to an operational interface.

3.4 Fermi Normal Coordinates: The Mathematical Implementation

The squeeze to an observer's worldline is mathematically realized by Fermi normal coordinates. In these coordinates, the metric reduces to Minkowski along the worldline ($g_{\mu\nu} = \eta_{\mu\nu}$) and the Christoffel symbols vanish there ($\Gamma^\mu_{\nu\rho} = 0$). Gravitational effects appear at second order in displacement from the worldline:

$$g_{00} \approx -1 - R_{0i0j}\, x^i x^j + \mathcal{O}(x^3). \tag{4}$$

All the physics is in the curvature. The coordinate freedom has been used up, "spent" on aligning the description with the observer's local measurements.

3.5 Payoff

In the tidal-first frame:

- The observable (arm-length change) appears directly as the primitive variable.
- Gauge artifacts are eliminated at the observer location.
- The "graviton" reappears not as a bulk perturbation but as a quantized curvature correlator $\langle E_{ij}(x)E_{kl}(x')\rangle$.

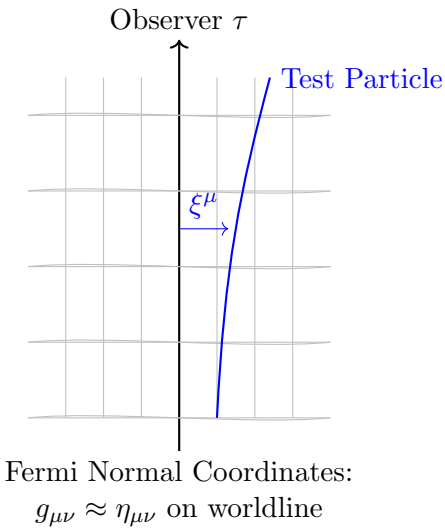

Figure 4: **Fermi Normal Coordinates.** The coordinate system is "squeezed" to the observer's worldline (vertical axis), where the metric is locally Minkowski. Curvature effects (grid distortion, geodesic deviation ξ^μ) appear only at second order in distance from the observer.

- The formulation makes explicit what LIGO analyses implicitly assume: that the relevant physics is the curvature along the detector's worldtube.

4 Example II: Boundary-First Gravity

Stakes: *This is why the BMS group and soft graviton theorems were discovered at null infinity rather than in the bulk—the structure was always there, but only visible from the boundary.*

4.1 The Particle Problem in Curved Spacetime

In Minkowski space, "particle" has a clean definition: it labels irreducible representations of the Poincaré group, or equivalently, positive-frequency modes of the field with respect to inertial time. In generic curved spacetimes, this definition fails. There is no global timelike Killing vector, no preferred mode decomposition, no invariant vacuum state.

The Unruh effect dramatizes the problem: the Minkowski vacuum, which an inertial observer sees as empty, appears to an accelerating observer as a thermal bath at temperature $T = a/2\pi$ (in natural units). The "particle content" depends on the observer's trajectory. This is not an artifact of approximation; it is an exact result.

If particle content is frame-dependent, where can we anchor a notion of gravitational radiation that supports particle-like language?

4.2 The Squeeze to Null Infinity

In asymptotically flat spacetimes, there is a natural boundary where the situation improves: future null infinity, denoted $\mathcal{I}^+$. This is the "end of the universe" for outgoing light rays and gravitational radiation. Distant observers, approximating inertial motion, collect radiation at $\mathcal{I}^+$.

At the boundary, several simplifications occur:

- The radiative degrees of freedom disentangle from the Coulombic (bound) degrees of freedom.
- The asymptotic metric approaches a universal form (Bondi gauge), exhausting most coordinate freedom.

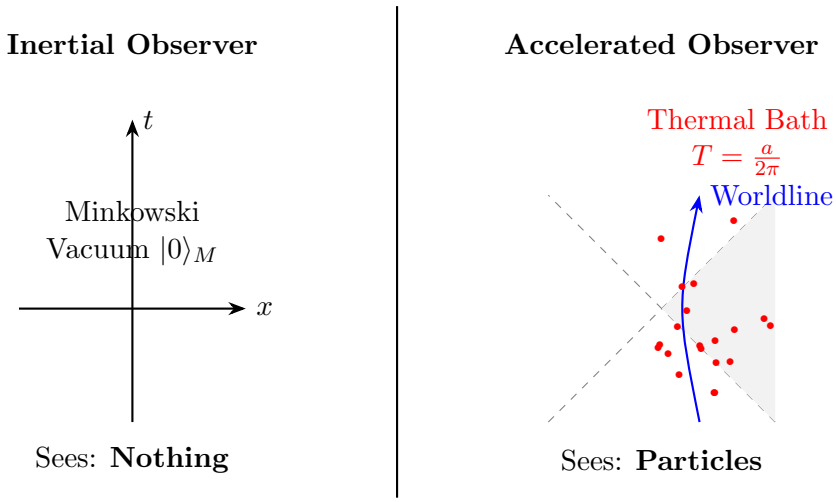

Figure 5: **The Unruh Effect.** Particle number is not an invariant. An inertial observer (left) sees a vacuum, while an accelerated observer (right) sees a thermal bath of particles. This demonstrates that "particle" is a frame-dependent concept in curved spacetime.

- A well-defined "news tensor" encodes the outgoing radiation flux.
- The Weyl scalar Ψ_4, which captures outgoing gravitational radiation, becomes the natural primitive.

From the boundary-first perspective, the bulk metric is a derived object reconstructed (in principle) from asymptotic data via characteristic evolution. The radiation *is* the boundary data.

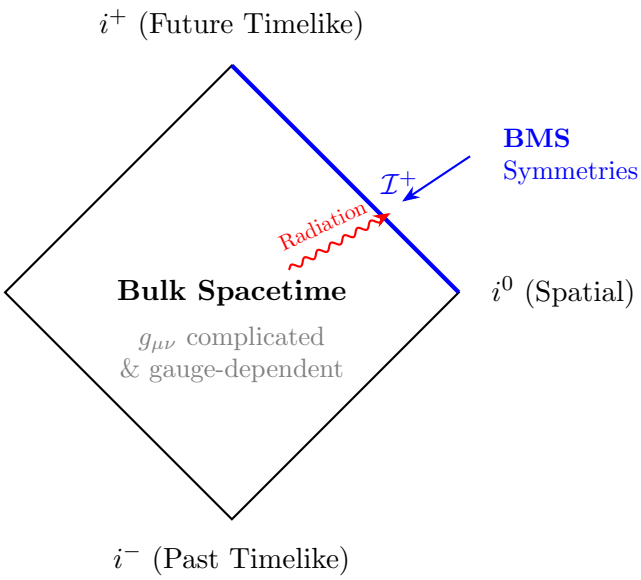

Figure 6: **Penrose Diagram of Asymptotically Flat Spacetime.** The bulk metric $g_{\mu\nu}$ is complicated and gauge-dependent. However, at Future Null Infinity ($\mathcal{I}^+$), the structure simplifies, revealing the BMS symmetries and defining radiation cleanly.

4.3 BMS Symmetries and Soft Theorems: Structure Revealed by the Squeeze

The boundary-first perspective revealed structure that was invisible in bulk analyses. The asymptotic symmetry group of gravity is not the Poincaré group but the much larger Bondi-Metzner-Sachs (BMS)

group, which includes angle-dependent translations called "supertranslations."

This larger symmetry was discovered in the 1960s by analyzing the structure of $\mathcal{I}^+$ directly. It was missed in bulk perturbation theory because the metric perturbation $h_{\mu\nu}$ does not cleanly encode the supertranslation charges—these are boundary observables.

More recently, the BMS symmetries were connected to:

- **Soft graviton theorems:** universal behavior of scattering amplitudes when one graviton has zero energy—a statement about $\mathcal{I}^+$ data.
- **Gravitational memory:** a permanent displacement of test masses after a gravitational wave passes—a DC component of the news tensor.
- **New conservation laws:** an infinite tower of charges associated with supertranslations, extending the usual Poincaré charges.

None of this structure is absent from the bulk formulation—general relativity is the same theory. But the bulk representation hid this structure inside gauge freedom and coordinate dependence. The squeeze to $\mathcal{I}^+$ made it manifest.

4.4 Payoff

The boundary-first frame:

- Provides a domain where particle language (mode counting, S-matrix) is well-defined.
- Revealed symmetry structure (BMS) invisible in canonical bulk variables.
- Connected asymptotic symmetries to soft theorems and memory effects—a unification achieved by representational shift, not new postulates.
- Suggests that the "graviton" is most cleanly defined as a boundary mode, not a bulk excitation.

5 Example III: The Dark Matter Problem (A Speculative Application)

Stakes: *We are not claiming to solve dark matter. We are claiming the representational question has been underexplored, and that one of the biggest mysteries in physics deserves a methodological audit.*

5.1 The Anomaly in Canonical Framing

Galactic rotation curves present a puzzle. In Newtonian gravity, orbital velocity should fall as $v \propto r^{-1/2}$ beyond the visible matter distribution. Observations show flat rotation curves: $v \approx$ const out to large radii.

The standard response has been ontological: add dark matter, a new substance that provides the missing gravitational pull. This hypothesis is well-motivated (cosmological evidence, structure formation) and may well be correct. We take no position on whether dark matter particles exist.

But notice the implicit representational commitments in the problem statement:

1. **Mass is primitive.** We count visible mass, find it insufficient, and infer invisible mass.
2. **The Newtonian regime is assumed.** Galactic dynamics are treated as a weak-field, slow-motion limit of GR.
3. **Acceleration is derived.** We compute expected acceleration from mass distribution via $\vec{a} = -\nabla\Phi$, then compare to observed velocities.

What happens if we invert this hierarchy?

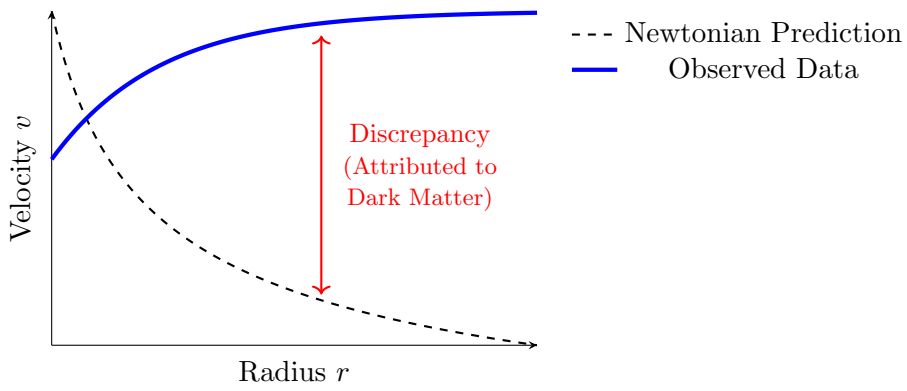

Figure 7: **Galactic Rotation Curves.** The discrepancy between the Newtonian prediction (dashed) and observed data (solid) is the primary evidence for dark matter. In the canonical frame, this is interpreted as missing mass.

5.2 An Acceleration-First Audit

Milgrom's MOND (Modified Newtonian Dynamics) is often framed as a modification of gravity. But it can also be read as a *representational observation*: galactic rotation curves exhibit a striking regularity when expressed in acceleration variables rather than mass variables.

The observation: the discrepancy between Newtonian prediction and observation appears below a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. Above this scale, dynamics are Newtonian. Below it, the effective gravitational acceleration approaches $\sqrt{a_N \cdot a_0}$ where a_N is the Newtonian prediction.

In mass-first coordinates, this looks like a fine-tuned coincidence—why should the dark matter distribution conspire to produce this regularity? In acceleration-first coordinates, it looks like a threshold phenomenon: something changes at a_0.

We are *not* endorsing MOND as a theory. We are noting that:

- The regularity is a fact about the data, independent of interpretation.
- This regularity is more naturally expressed in acceleration variables than in mass variables.
- A representational audit would ask: is there a formulation in which this regularity is manifest rather than anomalous?

5.3 A Curvature-First Audit

A different squeeze: instead of the Newtonian limit (weak field, slow motion), consider the full GR description with curvature as primitive.

In the Newtonian approximation, spacetime curvature is encoded entirely in the time-time component of the metric, and we recover $\nabla^2\Phi = 4\pi G\rho$. But this is an approximation that discards spatial curvature, gravitomagnetic effects, and the self-coupling of gravity.

The question is: in the regime where "dark matter" effects appear, are we sure the Newtonian approximation is valid? The accelerations are small ($a < a_0$), but the systems are also large (kpc scales) and the dynamics are non-linear over long timescales.

A representational audit would ask:

- What does the full curvature tensor look like in these systems?
- Are there degrees of freedom in the spatial curvature or gravitomagnetic sector that are discarded in the Newtonian limit?

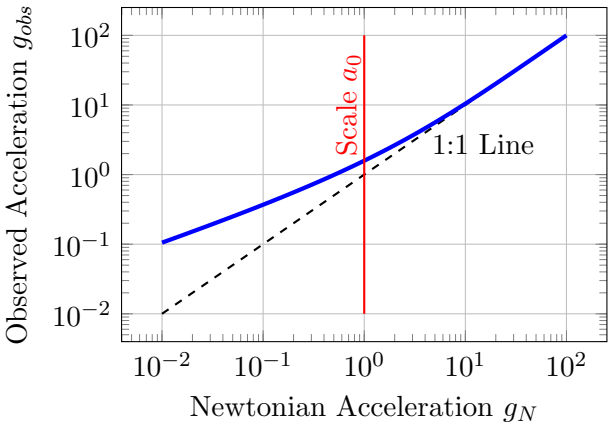

Figure 8: **The Radial Acceleration Relation (RAR).** When plotted in acceleration space, the data follows a tight regularity. The deviation from the 1:1 Newtonian line occurs specifically below a characteristic acceleration scale a_0, suggesting a threshold phenomenon.

- Does the Newtonian "squeeze" (weak-field, instantaneous action) hide structure that would be visible in a curvature-first analysis?

We do not have answers. We have *questions that arise from the method.*

5.4 What This Is and Isn't

This section is not:

- A claim that dark matter doesn't exist.
- An endorsement of MOND or any specific alternative gravity theory.
- A derivation showing that representational shift solves the problem.

This section is:

- An observation that the dark matter problem has been almost entirely approached through ontological addition (new particles) rather than representational audit.
- A demonstration that the method proposed in this paper generates non-trivial questions about the problem.
- A suggestion that the regularity seen in rotation-curve data may be telling us something about the right variables, not (only) about missing mass.

If even a partial resolution of the dark matter puzzle comes from recognizing representational artifacts, the return on this methodological investment is significant.

6 Historical Precedents

The stretch-and-squeeze method is not new. It is implicit in many of the most significant advances in physics. Extracting it as an explicit discipline is intended to make it available for systematic use rather than occasional inspiration.

The pattern is consistent: structure that was implicit (or invisible) in a canonical representation became manifest when the description was anchored to a different set of primitive observables. In no case was new physics added; existing physics was reorganized.

Table 2: Historical Cases of Representational Shift Revealing Hidden Structure

Phenomenon	**Original Representation**	**Squeezed Representation**	**What Became Visible**
Ether non-detection	Absolute rest frame; ether as medium	Observer-coupled measurements (rods, clocks)	Ether was unnecessary; simultaneity is relative
Gravity/inertia equivalence	Gravitational force as field on flat background	Free-fall frame; curvature as primitive	Gravity is geometry, not force
Dirac sea → antiparticles	Negative-energy electron sea	Time-reversed positive-energy particles	Antiparticles; simpler QED calculations
Unruh effect	Minkowski vacuum (inertial frame)	Rindler frame (accelerated observer)	Vacuum is thermal for accelerated observers
BMS symmetries	Bulk metric perturbation $h_{\mu\nu}$	Asymptotic data at $\mathcal{I}^+$	Supertranslations; soft theorems; memory
Constrained Hamiltonian systems	Lagrangian with redundant variables	Phase space with constraint surface	Clear separation of gauge and physical DoF

7 What Is Invariant Across Representations?

A reader committed to realism may ask: if particle content is frame-dependent (Unruh), if "force" is removable (equivalence principle), if primitive variables can be exchanged—what is the physics that remains constant?

This is a fair question, and the method does not evade it.

7.1 The Invariant Core

What survives all well-formed representational transformations:

1. **The algebraic structure of the observable algebra.** Different representations are different coordinatizations of the same algebra of observables. Commutation relations, correlation functions evaluated between physical states, and algebraic identities are invariant.

2. **Asymptotic observables.** S-matrix elements, scattering cross-sections, and memory effects are defined at boundaries where gauge freedom is exhausted. These are the "hard facts" that all representations must reproduce.

3. **Coincidences and correlations.** Whether two worldlines intersect, what a clock reads at an event, the ratio of two proper times—relational observables that do not depend on coordinates.

4. **Topological invariants.** Winding numbers, Chern classes, global charges—quantities defined without reference to local coordinates.

7.2 What Does Not Survive

- **Local particle number.** The Unruh effect shows this is observer-dependent.
- **Field values at coordinate points.** Gauge transformations change these.
- **Gravitational force at a point.** Removable by choice of reference frame.
- **Absolute simultaneity.** Killed by special relativity.

- **The distinction between "geometry" and "field on a background."** This is a representational choice, not a physical fact.

7.3 Philosophical Stance

This paper does not require commitment to a particular philosophy of science. The method is compatible with:

- **Structural realism:** The invariant structure (algebra, asymptotic data, relations) is what is real; representations are ways of coordinatizing it.
- **Operationalism:** Physical content is what can be measured; representations are bookkeeping.
- **Pragmatic realism:** Some representations are better than others for particular purposes; "real" is what plays the right functional role.

The method does not adjudicate among these positions. It does suggest that arguments about the "true" ontology of a theory (Is the metric fundamental? Are gravitons real?) may be less well-posed than they appear, if the quantities in question are representation-dependent.

8 Why This Is Counterintuitive

If representational audit is so valuable, why isn't it standard practice?

8.1 The Bias Toward Ontological Addition

When confronted with an anomaly, physicists are trained to propose explanations. An explanation, in the Hempelian tradition, subsumes the anomaly under a general law by introducing theoretical entities. Dark matter, the Higgs field, extra dimensions—these are all ontological additions that explain anomalies.

This training is valuable. But it creates a cognitive bias: adding structure feels like doing science; questioning coordinates feels like doing philosophy. The former is published in *Physical Review*; the latter in *Foundations of Physics*.

The result is an asymmetry of attention. Representational alternatives are explored by a small community working on "foundations," while the mainstream pursues increasingly elaborate ontological additions. Both are legitimate, but the portfolio is imbalanced.

8.2 Inherited Infrastructure

A canonical formulation becomes invisible through use. Students learn the metric formulation of GR, and the metric becomes "what gravity is." The fact that this is a representational choice, that tidal fields or connection forms could equally be primitive, recedes from view.

Textbooks reinforce this. They present *a* formulation, not the space of equivalent formulations. The exercises train facility with the canonical variables, not facility with moving between representations.

This is pedagogically necessary—one has to start somewhere. But it has a cost: physicists inherit a representation without inheriting the awareness that it *is* a representation.

8.3 The Rarity of Cross-Representational Fluency

Moving between representations is technically demanding. It requires not just knowing two formulations, but understanding the dictionary between them and maintaining a clear sense of which claims are representation-dependent.

This skill is rare because it is rarely taught systematically. The method proposed here is, in part, a pedagogical intervention: making representational audit an explicit step lowers the barrier to cross-representational work.

9 Limitations and Caveats

The stretch-and-squeeze method is not a magic wand.

9.1 Not All Elusiveness Is Representational

Some phenomena are genuinely hidden, not merely obscured by coordinate choice. Dark matter may exist as a new particle species, regardless of how we describe galactic dynamics. The cosmological constant may be a physical parameter, not an artifact of the coordinates we use for cosmology.

The method generates candidate questions, not guaranteed answers. The audit may conclude: "The canonical representation is fine; the phenomenon is genuinely new."

9.2 The Method Is Not Algorithmic

Choosing the "right" squeeze requires physical intuition about what observables are operationally relevant. There is no mechanical procedure for generating the optimal representation. The diagnostic checklist helps, but it does not replace judgment.

9.3 Foundational Work Is Slow

Even when a representational shift is productive, the payoff may take decades. The BMS group was discovered in the 1960s; its connection to soft theorems and memory effects became clear only in the 2010s. The method does not promise quick results.

9.4 Community Resistance

Proposing that a canonical formulation may be suboptimal is, implicitly, criticizing the accumulated practice of a field. This is sociologically difficult. The method is most likely to be taken up by researchers outside the mainstream, which may delay its influence.

10 Future Directions

Where might the method be applied next?

- **Quantum gravity:** Loop quantum gravity uses holonomy (connection) variables; string theory uses worldsheet (extended object) variables. Are these different squeezes of the same underlying structure? What invariants must both reproduce?
- **The black hole information paradox:** The paradox involves bulk (interior) and boundary (horizon, $\mathcal{I}^+$) descriptions. Are some versions of the paradox artifacts of mixing representations?
- **Dark energy and the cosmological constant:** The "unnaturalness" of Λ is computed in a particular effective field theory frame. Does the problem look different in a representation where Λ is not a bulk energy density but an asymptotic boundary condition?
- **Measurement in quantum mechanics:** The measurement problem arises in representations where the wave function is primitive and outcomes are derived. What does it look like in representations where outcomes (correlators, records) are primitive?

These are not research programs; they are questions that the method suggests. Whether they are fruitful is an empirical matter.

11 Conclusion

Einstein's greatest insights were not additions to physics but subtractions—the removal of representational artifacts (absolute rest, gravitational force) that had been mistaken for physical facts. This paper extracts that method and proposes it as an explicit discipline: *stretch-and-squeeze transformations.*

The method is simple in structure: expose gauge freedom (stretch), then compress to observer-coupled primitives (squeeze). The challenge lies in execution: identifying the right operational interface, maintaining invariants across the transformation, and resisting the cognitive pull toward ontological addition.

We illustrated the method with two cases where it has already paid off in gravitational physics: the tidal-field-first formulation that aligns with detector observables (LIGO), and the boundary-first formulation that revealed asymptotic structure (BMS symmetries, soft theorems). These are not new physics; they are existing physics reorganized to expose structure that was hidden in canonical variables.

We concluded with a speculative application to dark matter—not a solution, but an audit. The dark matter problem has been approached almost entirely through ontological addition. The representational question—whether the anomaly is partly an artifact of mass-first, Newtonian-limit framing—has been underexplored. We do not know the answer. We know the question deserves more attention.

The core claim is methodological: *when phenomena appear elusive, the first intervention should be representational, not ontological.* If even one major puzzle dissolves under this discipline, the return is enormous. And the history of physics, from ether to elevator, suggests the odds are good.

Acknowledgments. This paper offers no new physics, by design. Its claim is that representational hygiene is undersupplied in foundational physics, and that systematic attention to frame choice may dissolve puzzles currently approached through proliferating hypotheses.